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ABSTRACT

We theoretically design and experimentally demonstrate an invisibility concentrator, consisting of several truncated cylinders, for water waves based on a scattering cancellation method. The invisibility concentrator works by controlling the scattered waves from the target device. Our simulated and experimental results verify the concentration of waves and show the effective invisibility of the designed concentrator. This approach provides the possibility of simultaneously realizing wave concentration and an invisibility cloak, which has potential applications in energy harvesting.

The manipulation of waves has attracted the attention of many physicists due to its numerous potential applications. Transformation optics (TOs)[1–7] combined with metamaterials (MMs)[8–12] provide a powerful tool for manipulating waves that are propagating along randomly designed paths, as originally proposed for electromagnetic (EM) waves[1,2,11] and then extended to acoustic,[13–17] elastic,[18–20] and seismic waves.[21] Invisibility cloaks[22–25] and concentrators[26–31] are two typical wave-manipulation devices designed in accordance with TOs and MMs. Compared with TOs, the scattering cancellation method (SCM)[32] is an easier means of controlling waves and is typically used to fabricate invisibility cloaks.[32–38] As the name implies, this approach directly diminishes the scattered waves using plasmonic coats to cloak the object in waves, and the SCM has been successfully applied in EM[32,33,36,37] and acoustic[34–36,38] scenarios.

Water waves are a significant environmental factor that influences human activities in the oceans. As with other waves, the manipulation of water waves has two notable engineering applications, namely, invisibility cloaks and concentrators. Recently, TOs and MMs have been introduced to the field of water waves for the design of invisibility cloaks[39–44] and concentrators;[45] SCM has also been extended to water waves to cloak objects. Instead of plasmonic coats, the cloaking methods use designed topography[46–48] and surrounding bodies[49–56] to diminish the outgoing scattered waves from an object.

Generally, invisibility cloaks are used to protect objects by manipulating the waves so that they avoid the spaces in which the objects are located. In contrast, a concentrator is a device that focuses waves on the given space to enhance the energy density. In this sense, the concentrating and invisibility effects cannot be simultaneously implemented. Therefore, the fabrication of an invisibility concentrator is a novel topic in the field of wave manipulation. Previous research on this subject has involved introducing the Fabry–Perot (FP) resonance to TOs, resulting in an invisibility concentrator[29,57] shaped as a periodic metallic slit array for EM waves. This approach using FP resonance and TOs was later extended to water waves to design an annular invisibility concentrator[58] consisting of a periodic slit array with gradient depth profiles in the radial direction, which is suitable for shallow water.

In this letter, we describe the design and fabrication of an annular floating invisibility concentrator device for water waves. The proposed design relies on a completely different mechanism, namely, the direct manipulation of scattered waves. The fundamental concept of this invisibility concentrator is schematically illustrated in Fig. 1(a), where the expected propagation routes of incident waves, transmission waves, and scattered waves are demonstrated. This novel invisibility concentrator consists of several identical truncated cylinders arranged in a concentric circle. Instead of radiating to infinity, the scattered waves of the concentrator are focused inside

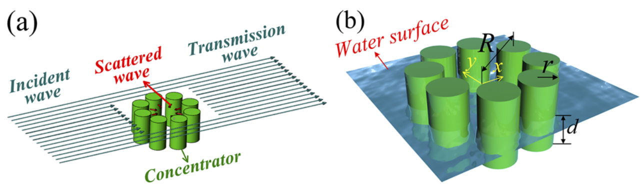

FIG. 1. Schematic diagram of the invisibility concentrator. (a) Fundamental concept. The incident waves propagate through the invisibility concentrator with full transmission because the scattered waves are concentrated inside. (b) Notation. Perspective view of the invisibility concentrator.

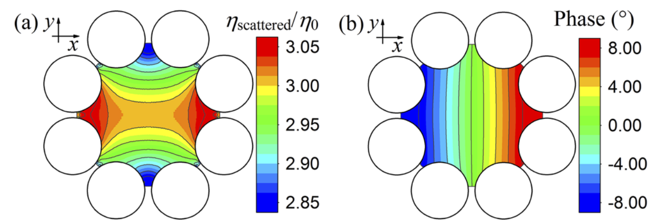

FIG. 2. Performance of the scattered waves inside the theoretically designed invisibility concentrator: (a) scattered wave amplitude ($\eta_{scattered}/\eta_0$); the scattered wave amplitude is symmetric about the y-axis. (b) Initial phase, which is antisymmetric about the y-axis.

the device; hence, the energy density inside the concentrator can be increased. As there are no scattered waves radiating to infinity, the concentrator is also cloaked.

Figure 1(b) schematically shows the notation of the invisibility concentrator in the perspective view. The invisibility concentrator is composed of eight fixed identical truncated cylinders of radius r and draft d placed in a concentric circle of radius R. The global coordinate system o-xyz is located at the center of the invisibility concentrator, the positive z-axis runs in the downward vertical direction, and the origin ($z = 0$) is placed on an undisturbed free surface.

Under the assumption of an incompressible and inviscid fluid with irrotational motion, Laplace's equation [Eq. (1)] is taken as the governing equation of the fluid field to establish this wave-structure interaction problem,

$$\nabla^2 \Phi(x, y, z; t) = 0, \tag{1}$$

where $\Phi(x, y, z; t)$ is the velocity potential of a field point (x, y, z) in the fluid space.

The velocity potential should satisfy the following linearized boundary conditions:

$$\begin{cases} \dfrac{\partial \Phi}{\partial t^2} + g\dfrac{\partial \Phi}{\partial z} = 0, & \text{at free surface} \\ \dfrac{\partial \Phi}{\partial n} = 0, & \text{at body surface} \\ \dfrac{\partial \Phi}{\partial z} = 0, & \text{at bottom,} \end{cases} \tag{2}$$

where g is the acceleration due to gravity.

A linear wave of amplitude η_0 over a flat bed of depth h traveling from infinity in a direction β relative to the positive x-axis is given by the following potential:

$$\varphi_I(P) = \frac{\cosh k_0(z-h)}{\cosh k_0 h} e^{-ik_0(x\cos\beta + y\sin\beta)}. \tag{3}$$

In this research, the deep-water condition is considered, while the waves are traveling from the negative x-axis. To solve this wave-structure problem, wave interaction theory (WIT)[59] is combined with the higher-order boundary element method (HOBEM).[60] The invisibility and concentration effects of the invisibility concentrator can be mathematically described by the attenuation coefficient α of the scattered wave energy as follows:

$$\alpha = \frac{E_s(r, d, R)}{E_0(r, d)}. \tag{4}$$

Here, $E_s(r, d, R)$[54] denotes the scattered wave energy of the whole invisibility concentrator, and $E_0(r, d)$ denotes the scattered wave energy of a single-element cylinder located alone in water waves.

For perfect invisibility, the physical quantity α should be zero. To model the invisibility concentrator, a real-coded genetic algorithm (RGA)[61] is adopted to minimize α by optimizing the values of r, d, and R [Fig. 1(b)].

Based on the above hydrodynamic methods (WIT, HOBEM) and the optimization algorithm (RGA), the invisibility concentrator is designed under the condition that the wave number k_0 is equal to 1.0 m^{-1} (frequency $f = 0.50$ Hz) in deep water. The initial search scope is $0.2\ \text{m} \le r \le 1.0$ m, $0.2\ \text{m} \le d \le 1.0$ m, and $0.8\ \text{m} \le R \le 4.0$ m. We obtained the dimensions of the invisibility concentrator and the corresponding attenuation coefficient (α) for the wave condition $k_0 = 1.0$ m^{-1} as follows: $r = 0.4285$ m, $d = 0.5471$ m, $R = 1.1269$ m, and $\alpha = 0.8968$. Although the value of α is not zero, we can find that the scattered wave energy (E_s) of the invisibility concentrator (consisting of eight truncated cylinders) is even less than that of a single

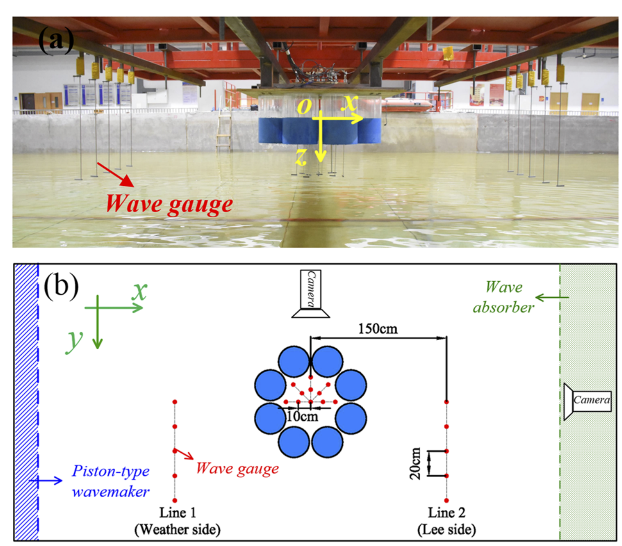

FIG. 3. Experimental apparatus. (a) Installation of the scale model and the wave gauges (side view). (b) Scale model arranged in a wave basin. 11 wave gauges are arranged inside the model, and 10 wave gauges are positioned at the two sides of the scale model. Two cameras are arranged at the side and the end of the basin to record the wave motion inside the scale model (top view).

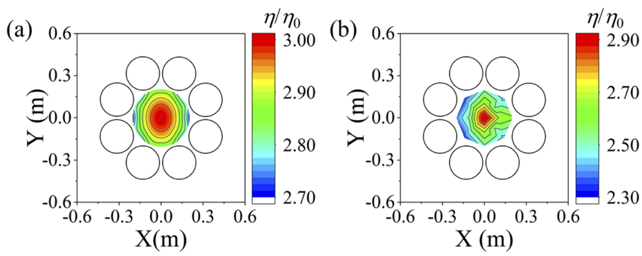

FIG. 4. Contour map of the wave amplitude (η/η_0) inside the invisibility concentrator for the wave frequency f = 0.91 Hz. (a) Numerical results of the wave amplitude in the range $R \leq 0.2$ m; the maximal values of the amplitude, η/η_0 = 2.91, can be found at the structural center. (b) Experimental results of the amplitude given by the measured results from 11 wave gauges, where the maximal wave amplitude (η/η_0) is 2.99.

cylinder (E_0), which means most scattered waves associated with the invisibility concentrator are focused inside the device. A resonance mode of scattered waves is found inside the invisibility concentrator (Fig. 2), which helps us interpret the working mechanism of this device.

A scale model [Fig. 3(a)] was fabricated to demonstrate the concentrating and invisibility effects of the device. The parameters of the installed scale model are r = 0.125 m, d = 0.181 m, and R = 0.353 m. Note that the dimensions of the scale model are not exactly equivalent to those of the theoretical device as we were limited by the physical installation conditions. We recalculated the attenuation coefficient (α) of the installed scale model under the incident wave f = 0.91 Hz and obtained a value of α = 1.1300, slightly larger than that of the theoretical device. Given the value of α in the installed scale model, we would expect the experimental device to exhibit effective concentrating and invisibility effects.

To demonstrate the concentrating and invisibility effects of the device, 21 capacitance-type wave gauges were installed at various locations, with 11 wave gauges inside the model and 10 wave gauges outside the model [Fig. 3(b)]. In addition to wave gauges, two cameras were arranged at the side and the end of the wave basin [Fig. 3(b)] to record the changes in the wave surface inside the device. Using the scale model, we conducted an experiment with a wavelength of λ = 1.878 m (frequency f = 0.91 Hz) and a wave height of H = 0.019 m and measured the wave amplitude η at the positions of the wave gauges.

First, we demonstrate the concentrating effects inside the device. Figure 4 shows the distribution of the dimensionless wave amplitude η/η_0 in the range $R \leq 0.2$ m inside the invisibility concentrator (here, η_0 is the amplitude of the incident waves). From the numerical results in Fig. 4(a), we find that all wave amplitudes inside the device have improved. The maximal values of the amplitude (η/η_0 = 2.91) can be found at the center of the invisibility concentrator. Figure 4(b) shows a contour map of the wave amplitude distribution given by the measured results from 11 wave gauges. The distribution law of the wave amplitude inside the invisibility concentrator accords with the numerical results, in which the maximal wave amplitude η/η_0 = 2.99 appears at the device center. Therefore, the concentrating effects of the designed invisibility concentrator have been verified.

To visualize the concentrating effects, we placed a circular cake-shaped float on the water surface without the concentrator [Fig. 5(a)] and inside the invisibility concentrator [Figs. 5(b) and 5(c)]. The float could only move along a pole with the heave motion of the water surface. Figure 5(a) shows the side view of the float amplitude without the concentrator, in which the amplitude is marked as an arrow. Figures 5(b) and 5(c) show screenshots of the float at the highest position inside the concentrator from the side view and the end view, respectively. Greater heave motion occurs in the float inside the concentrator, as shown in Figs. 5(b) and 5(c). Thus, our approach is of great significance for harvesting water wave energy.

Next, we numerically and experimentally investigated the invisibility effects of the scale model. For comparison, two single cylinders in water waves were also considered in the numerical simulations. One small single cylinder has the same radius as an element cylinder (r_s = 0.125 m) in the invisibility concentrator, whereas a big

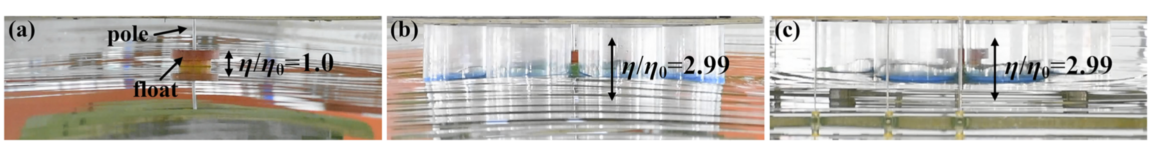

FIG. 5. Screenshots from the experimental video recorded by two cameras. (a) Single float in side view. The float inside the invisibility concentrator in (b) side view and (c) end view.

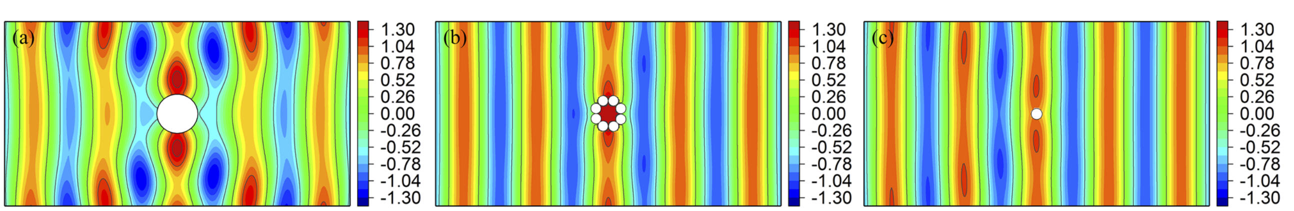

FIG. 6. Simulated field patterns of water waves for λ = 1.878 m (f = 0.91 Hz). Water free surface patterns for (a) big single cylinder, r_b = 0.478 m, (b) invisibility concentrator, and (c) small single cylinder, r_s = 0.125 m.

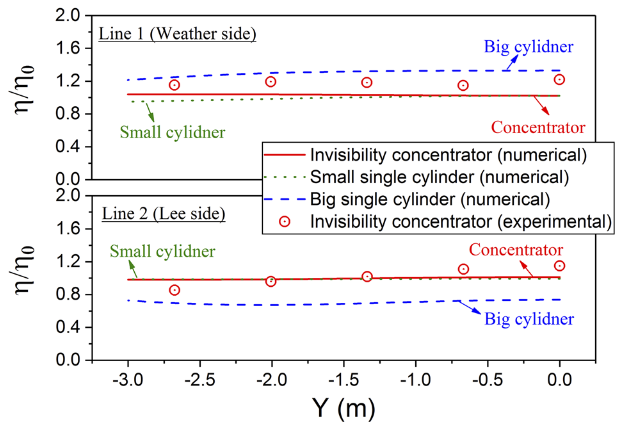

FIG. 7. Simulated and experimental wave amplitudes along two lines (line 1 and line 2). The simulated wave amplitudes along these two lines are presented for the invisibility concentrator, small single cylinder, and big single cylinder, and the measured wave amplitudes for the invisibility concentrator are shown for reference.

single cylinder has a radius of $r_b = 0.478$ m, which is the same as the circumradius $(R + r)$ of the invisibility concentrator arrangement. Both the invisibility concentrator and the two single cylinders have the same draft, i.e., $d = d_s = d_b = 0.181$ m.

Figures 6(a)–6(c) present the simulated free surface wave patterns of the big cylinder, the invisibility concentrator, and the small cylinder, respectively. As shown in Fig. 6(a), the free surface on both the weather side and the lee side is significantly disturbed in the presence of the big single cylinder. However, there is almost no interference to the environmental free surface in the presence of the invisibility concentrator [see Fig. 6(b)], though perfect invisibility is not realized. From Figs. 6(b) and 6(c), it is apparent that the incident waves propagate through the invisibility concentrator (consisting of eight small cylinders) just like they pass by one single small cylinder.

Figure 7 shows the measured amplitude (η/η_0) along two lines (Fig. 3) outside the invisibility concentrator, with the corresponding simulated wave amplitudes. For comparison, the numerical results of the big $(r = 0.478$ m) and small $(r = 0.125$ m) cylinders under the same wave conditions are also shown in Fig. 7. From the numerical results in Fig. 7, the simulated wave amplitudes of the invisibility concentrator are closer to those of the small cylinder along both lines. Regarding the wave amplitude in the presence of the invisibility concentrator, there is good agreement between the numerical results and the experimental results, which verifies the invisibility effects of the concentrator. From a hydrodynamic perspective, the invisibility effects can reduce the mean drift force acting on this floating device.[54] Overall, the results presented herein have verified that energy concentration and invisibility effects can be simultaneously realized by manipulating the scattered waves.

In conclusion, we have proposed a method for simultaneously realizing concentrating and invisibility effects in water waves by directly manipulating the scattered waves. Based on the proposed approach, we theoretically designed an invisibility concentrator by minimizing the attenuation coefficient (α) using an optimizing technique (combination of WIT, HOBEM, and RGA) and experimentally fabricated a scaled device in the laboratory. The experimental results agree well with the numerical simulations and demonstrate the efficiency of our approach in terms of energy concentrating and invisibility effects. We hope that our proposal can provide a new approach to harvesting water wave energy using a floating concentrator with a small mean drift force due to its invisibility effects. We believe that our design for an invisibility concentrator using the manipulation of scattered waves could also be introduced to other wave fields.

See the supplementary material for the movement of the float inside the invisibility concentrator.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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